Emanuele Iannone Ph.D. in Computer Science

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https://scholar.google.com/citations?user=tjOanfQAAAAJ
https://emaiannone.github.io

Personal Information

Full Name	Emanuele Iannone
Nationality	Italian
Birth	September 1, 1996

Work Experience

2023 – curr. **Postdoctoral Researcher**
Place: Institute of Software Security, Hamburg University of Technology, Germany
Start: November 1, 2023
👉 Employed full-time (as a *"Wissenschaftlicher Mitarbeiter"*) on the Horizon Europe project "Sec4AI4Sec" (Grant ID: 101120393). In charge of a work package in *automated vulnerability repair*.

Education

2020 – 2024 **Doctor of Philosophy (Ph.D.) in Computer Science**
Place: Software Engineering Lab, University of Salerno, Italy
Program Period: November 1, 2020 – October 31, 2023
Graduation Day: February 29, 2024
Thesis: *"There's Something About Vulnerabilities: Empirical Comprehension and Novel Automated Approaches"*, advised by Prof. Fabio Palomba.
Research Topic: Detection and assessment of software vulnerabilities in the context of software maintenance and evolution.

2018 – 2020 **Master of Science (M.Sc.) in Computer Science**
Place: University of Salerno, Italy
Thesis: *"Toward Automatic Exploit Generation of Known API Vulnerabilities"*, supervised by Prof. Fabio Palomba and Prof. Andrea De Lucia. This work resulted into a conference publication [C2].

2015 – 2018 **Bachelor of Science (B.Sc.) in Computer Science**
Place: University of Salerno, Italy
Thesis: *"Automated Refactoring of Energy-Related Code Smells of Android Applications"*, supervised by Prof. Andrea De Lucia. This work resulted into a conference publication [C1].

Research Visits

2022 **Visiting Ph.D. Student at University of Luxembourg (UniLu), Luxembourg**
Period: Oct 2022 – Dec 2022 (2 months)
Supervisor: Prof. Tegawendé F. Bissyande
Research Work: Synthesizing commit messages from bug reports using pre-trained transformers.

Visiting Ph.D. Student at Tampere University (TUNI), Finland
Period: May 2022 – Jun 2022 (1 month)
Supervisor: Prof. Davide Taibi
Research Work: Detecting vulnerability in open-source software using crowd-sourced information.

Publications

Journal Articles

[J7] T. Hinrichs, **E. Iannone**, and R. Scandariato, "Beyond prompting: The role of phrasing tasks in vulnerability prediction for java," *Cybersecurity*, vol. 8, no. 1, p. 111, 2025, ISSN: 2523-3246. DOI: 10.1186/s42400-025-00476-0.

[J6] T. Hinrichs, **E. Iannone**, T. Aladics, P. Hegedűs, A. De Lucia, F. Palomba, and R. Scandariato, "Back to the roots: Assessing mining techniques for java vulnerability-contributing commits," *ACM Trans. Softw. Eng. Methodol.*, 2025, ISSN: 1049-331X. DOI: 10.1145/3769105.

[J5] **E. Iannone**, G. Sellitto, E. Iaccarino, F. Ferrucci, A. De Lucia, and F. Palomba, "Early and realistic exploitability prediction of just-disclosed software vulnerabilities: How reliable can it be?" *ACM Trans. Softw. Eng. Methodol.*, vol. 33, no. 6, 2024, ISSN: 1049-331X. DOI: 10.1145/3654443.

[J4] **E. Iannone**, Z. Codabux, V. Lenarduzzi, A. De Lucia, and F. Palomba, "Rubbing salt in the wound? a large-scale investigation into the effects of refactoring on security," *Empirical Software Engineering*, vol. 28, no. 4, p. 89, 2023, ISSN: 1573-7616. DOI: 10.1007/s10664-023-10287-x.

[J3] F. Lomio, **E. Iannone**, A. De Lucia, F. Palomba, and V. Lenarduzzi, "Just-in-time software vulnerability detection: Are we there yet?" *Journal of Systems and Software*, p. 111 283, 2022, ISSN: 0164-1212. DOI: 10.1016/j.jss.2022.111283.

[J2] **E. Iannone**, R. Guadagni, F. Ferrucci, A. De Lucia, and F. Palomba, "The secret life of software vulnerabilities: A large-scale empirical study," *IEEE Transactions on Software Engineering*, vol. 49, no. 1, pp. 44–63, 2023. DOI: 10.1109/TSE.2022.3140868.

[J1] M. De Stefano, **E. Iannone**, F. Pecorelli, and D. A. Tamburri, "Impacts of software community patterns on process and product: An empirical study," *Science of Computer Programming*, vol. 214, p. 102 731, 2022, ISSN: 0167-6423. DOI: 10.1016/j.scico.2021.102731.

Conference Proceedings

[C9] **E. Iannone**, Q.-C. Bui, and R. Scandariato, "A match made in heaven? ai-driven matching of vulnerabilities and security unit tests," in *2026 IEEE/ACM 23rd International Conference on Mining Software Repositories (MSR)*, 2026.

[C8] Q.-C. Bui, **E. Iannone**, and R. Scandariato, "Vulterminator: Bringing back template-based automated repair for fixing java vulnerabilities," in *2026 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, 2026.

[C7] A. Cannavale, **E. Iannone**, G. Di Lillo, F. Palomba, and A. De Lucia, "The ground truth effect: Investigating szz variants in just-in-time vulnerability prediction," in *Software Engineering and Advanced Applications*, D. Taibi and D. Smite, Eds., Cham: Springer Nature Switzerland, 2025, pp. 317–326, ISBN: 978-3-032-04207-1. DOI: 10.1007/978-3-032-04207-1_21.

[C6] C. Tony, **E. Iannone**, and R. Scandariato, "Retrieve, refine, or both? using task-specific guidelines for secure python code generation," in *2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, 2025, pp. 368–379. DOI: 10.1109/ICSME64153.2025.00041.

[C5] **E. Iannone** and F. Palomba, "The phantom menace: Unmasking security issues in evolving software," in *2022 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, 2022, pp. 612–616. DOI: 10.1109/ICSME55016.2022.00085.

[C4] **E. Iannone**, M. De Stefano, F. Pecorelli, and A. De Lucia, "Predicting the energy consumption level of java classes in android apps: An exploratory analysis," in *2022 IEEE/ACM 9th International Conference on Mobile Software Engineering and Systems (MobileSoft)*, 2022, pp. 1–5. DOI: 10.1145/3524613.3527805.

[C3] G. Sellitto, **E. Iannone**, Z. Codabux, V. Lenarduzzi, A. De Lucia, F. Palomba, and F. Ferrucci, "Toward understanding the impact of refactoring on program comprehension," in *2022 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, 2022, pp. 731–742. DOI: 10.1109/SANER53432.2022.00090.

[C2] **E. Iannone**, D. Di Nucci, A. Sabetta, and A. De Lucia, "Toward automated exploit generation for known vulnerabilities in open-source libraries," in *2021 IEEE/ACM 29th International Conference on Program Comprehension (ICPC)*, 2021, pp. 396–400. DOI: 10.1109/ICPC52881.2021.00046.

[C1] **E. Iannone**, F. Pecorelli, D. Di Nucci, F. Palomba, and A. De Lucia, "Refactoring android-specific energy smells: A plugin for android studio," in *Proceedings of the 28th International Conference on Program Comprehension*, ser. ICPC '20, Seoul, Republic of Korea: Association for Computing Machinery, 2020, pp. 451–455, ISBN: 9781450379588. DOI: 10.1145/3387904.3389298.

Others

[O2] Q.-C. Bui, **E. Iannone**, M. Camporese, T. Hinrichs, C. Tony, L. Tóth, F. Palomba, P. Hegedűs, F. Massacci, and R. Scandariato, *A systematic literature review on automated exploit and security test generation*, 2025. arXiv: 2502.04953 [cs.CR].

[O1] **E. Iannone**, Q.-C. Bui, M. Isztin, B. Bogenfürst, G. Antal, P. Hegedűs, and R. Scandariato, *Vul4j+: A dataset of vulnerabilities for automated vulnerability repair*, https://doi.org/10.5281/zenodo.13752193, Online; accessed 12 March 2026, 2024.

Invited Talk and Guest Lectures

2025 Guest lecture on "**Dynamic Security Testing and Exploit Generation**", M.Sc. course *"International School"* of M.Sc. program *"Computer Security and Cloud Technologies"*, April 16, 2025, University of Salerno, Italy.

2023 Guest lecture on "**MSR for Vulnerability Prediction: Mining Vulnerability-Contributing Commits**", M.Sc. course *"Cybersecurity Data Science"*, May 31, 2023, TU Hamburg, Germany.

2022 Invited talk on "**MSR for Vulnerability Prediction: Lessons Learned, Challenges, and Recommendations**" at the *"Summer School on Security Testing and Verification 2022"*, September 22, 2022, KU Leuven, Belgium.

2021 Invited talk on "**Automated Test Case Generation: Toward Its Application in Exploit Generation for Known Vulnerabilities**" at the *"SAP Security Research Exchange Meeting"*, July 8, 2021, Online.

Reviewing Activities

Journals

I regularly serve as an invited reviewer for several Software Engineering-themed journals:

- **IEEE Transactions on Software Engineering (TSE)**
  Distinguished Reviewer in 2023
- **ACM Transactions on Software Engineering and Methodology (TOSEM)**
- **Springer's Empirical Software Engineering (EMSE)**
- **Elsevier's Journal of Systems and Software (JSS)**
- **IEEE Software Magazine**
- **Springer's Automated Software Engineering Journal (AUSE)**
- **Wiley's Journal of Software: Evolution and Process (JSEP)**
- **Elsevier's Science of Computer Programming (SCICO)**

So far, I have reviewed over **55** journal papers, including revised submissions.

Reviewing Activities (continued)

Conference Program Committee

I regularly serve in the program committee for several Software Engineering conferences and workshops across various tracks. So far, I have reviewed over **90** conference and workshop papers.

2026 **[ICSME 2026]** *42nd Int. Conf. on Software Maintenance and Evolution*, Research + Industry + Replication and Negative Results Tracks.

[DeMeSSAI 2026] *5th Int. Workshop on Designing and Measuring Security in Systems with AI*.

[ICST 2026] *19th IEEE Int. Conf. on Software Testing, Verification and Validation*, Testing Tools and Data Showcase Track.

[FSE 2026] *33rd ACM Int. Conf. on the Foundations of Software Engineering*, Student Research Competition Track.

[EASE 2026] *30th Int. Conf. on Evaluation and Assessment in Software Engineering*, Research Track.

2025 **[SQA4AI 2026]** *3rd Int. Workshop on Software Quality Assurance for Artificial Intelligence*.

[MSR 2026] *22nd Int. Conf. on Mining Software Repositories*, Data and Tool Showcase Track.

[SANER 2026] *33rd IEEE Int. Conf. on Software Analysis, Evolution and Reengineering*, Research Track.

[ASE 2025] *40th IEEE/ACM Int. Conf. on Automated Software*, New Ideas and Emerging Results Track.

[EASE 2025] *29th Int. Conf. on Evaluation and Assessment in Software Engineering*, Posters and Vision Track.

[ICSME 2025] *41st Int. Conf. on Software Maintenance and Evolution*, Research + Tool Demonstration Tracks.

Distinguished Reviewer Award

[DeMeSSAI'25] *4th Int. Workshop on Designing and Measuring Security in Systems with AI*.

[FSE 2025] *32nd ACM Int. Conf. on the Foundations of Software Engineering*, Ideas, Visions and Reflections Track.

2024 **[SQA4AI 2025]** *2nd Int. Workshop on Software Quality Assurance for Artificial Intelligence*.

[SANER 2025] *32nd IEEE Int. Conf. on Software Analysis, Evolution and Reengineering*.

[ICSME 2024] *40th Int. Conf. on Software Maintenance and Evolution*, New Ideas and Emerging Results Track.

[ASE 2024] *39th IEEE/ACM Int. Conf. on Automated Software Engineering*, New Ideas and Emerging Results Track.

[MSR 2025] *21st Int. Conf. on Mining Software Repositories*, Data and Tool Showcase Track

[SANER 2024] *31st IEEE Int. Conf. on Software Analysis, Evolution and Reengineering*, Research Track.

2023 **[SSBSE 2023]** *15th Symposium on Search-Based Software Engineering*, RENE/NIER Track.

[MOBILESoft 2023] *10th Int. Conf. on Mobile Software Engineering and Systems*, Research Track.

[MSR 2023] *20th Int. Conf. on Mining Software Repositories*, Research Track ("Junior" PC Member).

2022 **[CSCW 2023]** *26th ACM Conf. on Computer-Supported Cooperative Work & Social Computing* (Invited Reviewer).

[SSBSE 2022] *14th Symposium on Search-Based Software Engineering*, RENE/NIER Track

[ASE 2022] *37th IEEE/ACM Int. Conf. on Automated Software Engineering*, New Ideas and Emerging Results Track.

[MSR 2022] *19th Int. Conf. on Mining Software Repositories*, Research Track ("Shadow" PC Member).

[SQA4AI 2022] *1st Int. Workshop on Software Quality Assurance for Artificial Intelligence*.

[CSCW 2022] *25th ACM Conf. on Computer-Supported Cooperative Work & Social Computing* (Invited Reviewer).

Reviewing Activities (continued)

2021 **[ICSEA 2021]** *16th Int. Conf. on Software Engineering Advances*.

Event Organization and Community Services

2025 **Guest Editor** for *EMSE's Special Issue* on Security Testing for Complex Software Systems (SECUTE).

2024 **Social Media Editor** for *Science of Computer Programming* journal, ongoing since March 4, 2024.

2023 **Member of the Local Arrangement Team** for SATToSE 2023 (*15th Seminar on Advanced Techniques & Tools for Software Evolution*) and ISSSE 2023 (*16th International Summer School on Software Engineering*), Fisciano, Italy.

2022 **Student Volunteer** for ICSE 2022 (*44th IEEE/ACM Int. Conf. on Software Engineering*) and Co-located Events (held remotely).

2021 **Publicity Co-Chair** for ICPC 2021 (*29th IEEE/ACM Int. Conf. on Program Comprehension*).

Miscellaneous

Awards and Achievements

2025 **Distinguished Reviewer Award at ICSME 2025** (*41st IEEE Int. Conf. on Software Maintenance and Evolution*)

2023 **IEEE Transactions on Software Engineering (TSE) Distinguished Reviewer** for Year 2023

2022 **IEEE-CS TCSE Distinguished Paper Award** for *"Toward Understanding The Impact of Refactoring on Program Comprehension"* published at SANER 2022 [C3].

Skills

Languages	Italian (Native), English (Fluent)
Technology	Java, Python, C, R, HTML, CSS, JavaScript, Angular, Ionic, Capacitor, MySQL, Neo4J, Spring Boot, Android, Git, Bash, Docker, LaTeX